

The General Theory of Demand and Supply of Monetary Assets

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Abstract

This paper develops a general theory of the demand and supply of monetary assets by treating money as a competitive economic industry rather than assuming a particular monetary instrument to be the economically privileged numeraire. The framework encompasses fiat currencies, cryptocurrencies, precious metals, industrial commodities with monetary functions, and other assets capable of providing monetary services. The central object is the general equilibrium allocation of scarce wealth among competing monetary technologies.

The theory distinguishes institutional scarcity, algorithmic scarcity, and physical scarcity; protocol supply from effective market supply; monetary demand from non-monetary commodity demand; transactional, precautionary, speculative, investment, collateral, and store-of-value demand; and nominal prices from economically meaningful relative prices. It establishes a multi-asset monetary equilibrium in which fiat currencies, cryptocurrencies, and commodity monies are simultaneously substitutes and, under appropriate conditions, complements.

The framework integrates monetary economics, finance, general equilibrium theory, industrial organization, network economics, behavioral economics, statistics, information theory, computer science, artificial intelligence, political economy, international relations, geopolitics, diplomacy, welfare economics, and warfare economics. Particular attention is given to network effects, expectations, credibility, liquidity, institutional power, sanctions, warfare, strategic reserves, technological infrastructure, and the physical and computational constraints underlying alternative monetary systems.

The resulting theory implies that a cryptocurrency price quoted in fiat is not an absolute economic price: it is one relative price within a larger monetary price system. The same principle applies to gold, silver, copper, platinum, and fiat currencies themselves. The fundamental economic problem is therefore the determination of relative monetary purchasing powers and the equilibrium allocation of wealth among competing monetary assets.

The paper ends with “The End”

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1 Introduction

Money is frequently modeled as though its identity were given in advance. A conventional monetary model begins with a national fiat currency, while a cryptocurrency model often begins with a cryptocurrency whose price is quoted in that fiat currency. A commodity-money model may instead begin with gold or silver. Each approach is useful, but each risks obscuring a deeper economic fact:

different monetary assets compete with one another.

A household, corporation, bank, institutional investor, sovereign, or military organization may simultaneously hold fiat currency, cryptocurrency, gold, silver, other commodities, bank deposits, securities, and inventories that provide monetary or quasi-monetary services.

Consequently, the economically fundamental question is not simply

“What is the price of Bitcoin in dollars?”

but rather

“What determines the relative purchasing powers of competing monetary assets?”

This paper develops a general framework for answering that question.

Let $\mathcal{M}$ denote the set of monetary assets. We write

$$\mathcal{M} = \mathcal{F} \cup \mathcal{C} \cup \mathcal{G} \cup \mathcal{O},$$

where $\mathcal{F}$ denotes fiat currencies, $\mathcal{C}$ denotes cryptocurrencies, $\mathcal{G}$ denotes commodity monies, and $\mathcal{O}$ denotes other monetary or quasi-monetary assets.

The theory begins from the principle that every monetary asset has a demand and a supply, and that equilibrium is determined jointly.

2 The Monetary Asset as an Economic Object

Definition 1 (Monetary Asset). *A monetary asset is an asset that provides one or more monetary services, including medium-of-exchange, unit-of-account, store-of-value, settlement, collateral, liquidity, or purchasing-power-preservation services.*

The definition deliberately avoids requiring legal tender.

A fiat currency can possess monetary utility because of taxation, legal recognition, banking infrastructure, and network effects. A cryptocurrency can possess monetary utility because of decentralized settlement, programmability, scarcity rules, and network adoption. Gold can possess monetary utility because of scarcity, durability, divisibility, portability, historical credibility, and resistance to certain forms of political interference.

Hence monetary status is not necessarily binary.

Define the monetary-service vector of asset j by

$$\mathbf{m}_j = \begin{pmatrix} T_j \\ L_j \\ S_j \\ C_j \\ N_j \\ A_j \end{pmatrix},$$

where:

- T_j is transaction utility;
- L_j is liquidity;
- S_j is store-of-value utility;
- C_j is collateral utility;
- N_j is network utility;
- A_j is accounting or unit-of-account utility.

The economic value of an asset is therefore not necessarily exhausted by its physical or financial payoff.

3 Three Fundamental Monetary Technologies

The theory distinguishes three major scarcity technologies.

3.1 Fiat Money

Fiat supply is principally institutionally determined:

$$M_F^S = M_F^S(\text{central-bank policy, bank credit, fiscal policy, regulation}).$$

Its scarcity is therefore institutional.

3.2 Cryptocurrency

Cryptocurrency supply is determined by protocol and network rules:

$$M_C^S(t) = M_C(0) + \int_0^t [I_C(\tau) - B_C(\tau)] d\tau,$$

where I_C represents issuance and B_C represents effective burning or destruction.

Its scarcity may therefore be called algorithmic or protocol-based scarcity.

3.3 Commodity Money

Commodity-money supply is constrained by physical resources:

$$M_G^S(t) = M_G(0) + \int_0^t [X_G(\tau) + R_G(\tau) - D_G(\tau)] d\tau,$$

where X_G is extraction, R_G is recycling, and D_G is physical destruction or irreversible dissipation.

Its scarcity is ultimately physical and technological.

Thus:

Institutional Scarcity $\leftrightarrow$ Algorithmic Scarcity $\leftrightarrow$ Physical Scarcity.
--

This distinction is central to the theory.

4 Physical Commodity Money

Gold, silver, copper, platinum, and related commodities occupy a special position because they can have simultaneous monetary and non-monetary uses.

For commodity g ,

$$Q_g^D = Q_g^M + Q_g^I + Q_g^J + Q_g^S,$$

where:

- Q_g^M is monetary demand;
- Q_g^I is industrial demand;
- Q_g^J is jewelry or cultural demand;
- Q_g^S is speculative and investment demand.

The equilibrium condition is

$$Q_g^D(P_g, \mathbf{P}, \mathbf{X}) = Q_g^S(P_g, \mathbf{Z}).$$

A monetary shock can therefore influence a commodity price through one channel while an industrial shock influences it through another.

This is an important difference between physical monetary commodities and purely digital monetary assets.

5 Cryptocurrency Supply

A cryptocurrency has several distinct notions of supply.

Definition 2 (Protocol Supply). *Protocol supply is the quantity of units permitted by the cryptocurrency's consensus and issuance rules.*

Definition 3 (Circulating Supply). *Circulating supply is the quantity technically outstanding and potentially transferable.*

Definition 4 (Effective Market Supply). *Effective market supply is the quantity actually offered or economically available for exchange at a particular price and time.*

These quantities need not coincide.

Let

$$S_C^P(t)$$

denote protocol supply. Effective market supply can be written as

$$S_C^M(P, t) = S_C^P(t) - H_C(P, t) - L_C(P, t) + U_C(P, t),$$

where:

- H_C is cryptocurrency withheld by long-term holders;
- L_C is cryptocurrency locked in staking, bridges, contracts, or other mechanisms;
- U_C is previously illiquid cryptocurrency becoming available.

Hence a cryptocurrency may possess a fixed or predetermined terminal supply while still having an elastic effective supply.

6 Fiat Supply

Fiat currency is produced through a hierarchy involving the monetary authority, commercial banking system, fiscal authority, and payment infrastructure.

A stylized monetary base equation is

$$B_t = B_{t-1} + \Delta A_t - \Delta L_t,$$

where ΔA_t represents changes in monetary-authority assets and ΔL_t represents changes in relevant liabilities or sterilization.

Broad money can then be represented schematically as

$$M_t = \mu_t B_t,$$

where μ_t is an endogenous money multiplier or balance-sheet expansion factor.

Unlike a simple textbook multiplier, μ_t may depend on capital requirements, liquidity constraints, risk appetite, regulation, endogenous credit demand, and banking technology.

7 Demand for Monetary Assets

For monetary asset j , define utility as

$$V_j = V_j(R_j, \sigma_j, L_j, T_j, N_j, S_j, I_j, K_j, G_j, \mathbf{X}),$$

where:

- R_j is expected return;
- σ_j is risk;
- L_j is liquidity;

- T_j is transaction utility;
- N_j is network utility;
- S_j is scarcity;
- I_j is institutional credibility;
- K_j is collateral utility;
- G_j represents geopolitical characteristics;
- $\mathbf{X}$ contains other state variables.

A representative portfolio allocation may be written as

$$s_j = \frac{\exp(V_j)}{\sum_{k \in \mathcal{M}} \exp(V_k)}.$$

Consequently,

$$\sum_{j \in \mathcal{M}} s_j = 1.$$

This is the fundamental portfolio-competition constraint.

8 Relative Prices

Let P_i and P_j denote the prices of monetary assets i and j in an arbitrary reference unit. Their economically meaningful relative price is

$$P_{ij} = \frac{P_i}{P_j}.$$

For cryptocurrency C , fiat F , and commodity money G ,

$$P_{C/F} = \frac{P_C}{P_F},$$

$$P_{G/F} = \frac{P_G}{P_F},$$

and

$$P_{C/G} = \frac{P_C}{P_G}.$$

They satisfy

$$P_{C/F} = P_{C/G} P_{G/F}.$$

Equivalently,

$$P_{C/F} P_{G/C} P_{F/G} = 1.$$

This is the triangular arbitrage condition.

Proposition 1 (Relative-Price Principle). *The quotation of a monetary asset in fiat units does not make fiat an economically neutral object. It specifies one relative price within a multi-asset monetary system.*

Proof. Let P_C and P_F denote prices in an arbitrary external unit. The quoted crypto price in fiat is $P_{C/F} = P_C/P_F$. A change in $P_{C/F}$ can arise from a change in P_C , a change in P_F , or both. Hence the observed fiat price does not identify a unique change in the intrinsic economic attractiveness of cryptocurrency. Only relative purchasing power is economically meaningful. $\square$ $\square$

9 Fiat–Crypto Competition

Let s_C and s_F denote wealth shares invested in cryptocurrency and fiat.
The budget constraint is

$$W = P_F M_F + P_C M_C.$$

The relative attractiveness of cryptocurrency may be represented by

$$\Delta V_{CF} = V_C - V_F.$$

Then

$$s_C = \frac{\exp(V_C)}{\exp(V_C) + \exp(V_F)}.$$

Therefore,

$$\frac{\partial s_C}{\partial V_C} > 0, \quad \frac{\partial s_C}{\partial V_F} < 0.$$

Fiat inflation can enter the system through

$$V_F = V_F(\pi_F^e, \text{credibility, liquidity, legal status, } \dots).$$

If

$$\frac{\partial V_F}{\partial \pi_F^e} < 0,$$

then

$$\frac{\partial s_C}{\partial \pi_F^e} > 0$$

provided that cryptocurrency utility is held constant.

This is a substitution result, not a claim that inflation mechanically determines cryptocurrency prices.

10 Crypto–Commodity Competition

Cryptocurrency also competes with commodity money.

Define

$$\Delta V_{CG} = V_C - V_G.$$

Then the demand share for cryptocurrency relative to commodity money depends upon

$$R_C - R_G, \quad L_C - L_G, \quad S_C - S_G, \quad N_C - N_G, \quad I_C - I_G.$$

A technological improvement increasing cryptocurrency transaction utility can therefore reduce demand for gold, silver, or another monetary commodity even if the commodity's industrial demand remains unchanged.

Conversely, a geopolitical crisis that increases the perceived physical security or institutional neutrality of gold can increase V_G and reduce the portfolio shares of competing monetary assets.

11 The Three-Way Monetary Equilibrium

Consider three monetary classes:

$$\mathcal{M} = \{F, C, G\}.$$

Total monetary wealth satisfies

$$W = P_F M_F + P_C M_C + P_G M_G.$$

The portfolio shares satisfy

$$s_F + s_C + s_G = 1.$$

Demand functions are

$$M_F^D = D_F(P_{C/F}, P_{G/F}, \mathbf{X}),$$

$$M_C^D = D_C(P_{C/F}, P_{C/G}, \mathbf{X}),$$

and

$$M_G^D = D_G(P_{G/F}, P_{C/G}, \mathbf{X}).$$

Equilibrium requires

$$M_F^D = M_F^S,$$

$$M_C^D = M_C^S,$$

and

$$M_G^D = M_G^S.$$

The exchange rates are simultaneously determined by the system.

12 General n -Asset Monetary Equilibrium

Let

$$\mathcal{M} = \{1, 2, \dots, n\}.$$

For each asset i ,

$$Q_i^D = D_i(P_{i1}, \dots, P_{in}, \mathbf{X}),$$

and

$$Q_i^S = S_i(P_{i1}, \dots, P_{in}, \mathbf{Z}).$$

Equilibrium is defined by

$$\boxed{D_i(\mathbf{P}, \mathbf{X}) = S_i(\mathbf{P}, \mathbf{Z}), \quad i = 1, \dots, n.}$$

Because only relative prices matter, one price can be normalized:

$$P_n = 1.$$

The equilibrium then determines

$$\mathbf{P}^* = (P_1^*, \dots, P_{n-1}^*).$$

The normalization is mathematical rather than ontological.

13 Network Effects

Cryptocurrency introduces a powerful endogenous network effect.

Let N_i be the number of economically active users.

Define

$$U_i(N_i) = a_i N_i^{\alpha_i}, \quad a_i > 0.$$

Then

$$\frac{\partial U_i}{\partial N_i} = a_i \alpha_i N_i^{\alpha_i - 1}.$$

If

$$\alpha_i > 1,$$

the utility effect is superlinear.

Demand may therefore satisfy

$$Q_i^D = D_i(P_i, N_i, E_t[N_{i,t+1}], \dots).$$

This creates a feedback mechanism:

$$\text{adoption} \rightarrow \text{network utility} \rightarrow \text{demand} \rightarrow \text{investment} \rightarrow \text{adoption}.$$

Such feedback can generate multiple equilibria.

14 Behavioral Economics and Psychology

Monetary demand is not necessarily determined solely by rational expected returns.

Let an agent's perceived utility be

$$\tilde{V}_i = V_i + B_i + A_i + H_i,$$

where:

- B_i represents behavioral biases;
- A_i represents attention and salience;
- H_i represents heterogeneous beliefs.

Relevant behavioral mechanisms include:

- loss aversion;
- extrapolation;
- recency bias;
- herd behavior;
- ambiguity aversion;
- overconfidence;
- narrative contagion;
- status consumption;
- fear of missing out;
- distrust of institutions.

A representative demand equation can therefore be written as

$$Q_i^D = D_i(P_i, E[P_{i,t+1}], \text{Var}(P_{i,t+1}), \text{attention, beliefs, network size}).$$

This provides a bridge between standard economics and behavioral finance.

15 Psychiatric and Cognitive Considerations

The theory does not require pathological behavior to explain market volatility. Nevertheless, psychological states can affect aggregate risk perception, attention, impulsivity, and susceptibility to social contagion.

Let

$$\mathcal{B}_t$$

represent the aggregate distribution of behavioral states.

Then

$$Q_i^D = D_i(P_i, \mathbf{X}_t, \mathcal{B}_t).$$

A scientifically responsible model should treat psychiatric variables as latent or population-level behavioral states rather than diagnosing individual market participants.

The relevant economic mechanism is therefore:

$$\text{cognition} \rightarrow \text{belief formation} \rightarrow \text{portfolio choice} \rightarrow \text{market demand}.$$

16 Finance and Asset Pricing

A monetary asset can provide both monetary services and financial returns.

For asset i ,

$$R_{i,t+1} = \frac{P_{i,t+1} + D_{i,t+1}}{P_{i,t}} - 1,$$

where $D_{i,t+1}$ represents dividends, staking rewards, interest, convenience yields, or analogous benefits. The expected excess return relative to fiat is

$$E_t[R_{i,t+1} - R_{F,t+1}].$$

A generalized stochastic discount factor condition is

$$E_t[m_{t+1}R_{i,t+1}] = 1.$$

For a non-income-producing monetary asset, monetary services enter utility rather than appearing as conventional dividends.

This provides a connection between monetary economics and modern asset pricing.

17 Commodity Convenience Yield

Physical monetary commodities possess a potential convenience yield.

Let

$$CY_G$$

denote the convenience yield of holding physical commodity G .

A simplified pricing relationship is

$$R_G \approx r + c - CY_G,$$

where r is the financing cost and c represents storage and carrying costs.

For monetary commodities, CY_G may include:

- liquidity;
- scarcity;
- crisis insurance;
- collateral utility;

- settlement utility;
- strategic-reserve value.

This is one reason why physical commodities cannot be reduced to ordinary industrial inputs.

18 Physics and Physical Constraints

Commodity monetary assets obey physical conservation constraints that digital assets do not.

For a physical stock G_t ,

$$G_{t+1} = G_t + X_t + R_t - C_t - D_t.$$

Here C_t represents consumption and D_t physical losses.

The extraction process is constrained by energy and technology.

Let

$$E_t \geq E(G_t, X_t)$$

represent the energy requirement for extraction.

Consequently, physical monetary scarcity is partly an energetic constraint.

By contrast, cryptocurrency issuance can be computationally constrained:

$$E_t^{comp} = E_t^{hardware} + E_t^{electricity} + E_t^{network}.$$

Thus the supply technologies of monetary assets are connected to different physical resource systems.

19 Chemistry and Material Properties

The chemical properties of monetary commodities affect durability, corrosion resistance, divisibility, purification, recycling, and storage.

Let a commodity's physical monetary quality be represented by

$$\Phi_G = \Phi(\text{chemical stability, purity, density, corrosion resistance, malleability}).$$

Gold's chemical stability, for example, differs substantially from copper's industrial characteristics.

Thus the economic supply curve of commodity money cannot be completely separated from chemistry and materials science.

20 Biology and Evolution

Monetary institutions can be interpreted as evolved coordination mechanisms.

Let $p_i(t)$ denote the fraction of a population using monetary technology i .

A replicator-type dynamic is

$$\dot{p}_i = p_i \left(V_i - \sum_j p_j V_j \right).$$

If V_i exceeds the population-weighted average utility, adoption increases.

This yields a biological interpretation of monetary competition:

$$\text{variation} \rightarrow \text{selection} \rightarrow \text{adoption} \rightarrow \text{institutional persistence}.$$

The analogy is behavioral and evolutionary rather than a claim that monetary institutions are literally biological organisms.

21 Information Theory

Monetary assets operate under uncertainty.

Let the probability distribution of future states be

$$p(s).$$

The Shannon entropy is

$$H(S) = - \sum_s p(s) \log p(s).$$

Assets that reduce uncertainty about future purchasing power may possess additional monetary utility. Define information utility as

$$I_i = -\lambda H_i, \quad \lambda > 0.$$

Credible monetary rules can therefore generate economic value through uncertainty reduction. This applies differently to:

- central-bank rules;
- constitutional monetary rules;
- cryptocurrency consensus protocols;
- physical scarcity;
- strategic commodity reserves.

22 Computer Science and Protocol Economics

Cryptocurrencies are monetary systems implemented partly through algorithms.

A protocol can be represented as

$$\mathcal{P} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{C}, \mathcal{R}),$$

where:

- $\mathcal{S}$ is the state space;
- $\mathcal{A}$ is the action space;
- $\mathcal{T}$ is the state-transition rule;
- $\mathcal{C}$ is the consensus mechanism;
- $\mathcal{R}$ is the reward and issuance rule.

The protocol determines a state transition

$$s_{t+1} = T(s_t, a_t, \omega_t).$$

This creates an unusual form of institutional technology: rules can be partially encoded into executable software.

23 Algorithmic Representation

A computational equilibrium algorithm can be written as follows.

INPUT:

```
monetary assets M
initial prices P
supply functions S_i
demand functions D_i
network states N_i
macroeconomic state X
geopolitical state G
```

REPEAT:

```
FOR each monetary asset i:
  compute utility V_i
  compute demand Q_i^D
  compute effective supply Q_i^S
  compute excess demand Z_i = Q_i^D - Q_i^S
```

UPDATE relative prices:

```
P_i <- P_i * exp(lambda * Z_i)
```

UPDATE network states:

```
N_i <- NetworkUpdate(N_i, Q_i^D)
```

UPDATE expectations:

```
E_i <- LearningRule(P_i, history)
```

UPDATE geopolitical and institutional states:

```
G <- StateTransition(G)
```

UNTIL:

```
max_i |Z_i| < tolerance
```

OUTPUT:

```
equilibrium relative prices
equilibrium quantities
portfolio shares
network states
monetary dominance indicators
```

This algorithm is not a claim that markets literally execute the pseudocode. It is a computational representation of the equilibrium mechanism.

24 Statistics and Econometrics

Let $P_{i,t}$ denote the observed price of monetary asset i .

Returns are

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1.$$

Log returns are

$$\ell_{i,t} = \log P_{i,t} - \log P_{i,t-1}.$$

A demand system may be estimated as

$$Q_{i,t}^D = \alpha_i + \beta_i P_{i,t} + \sum_{j \neq i} \gamma_{ij} P_{j,t} + \delta_i X_t + \varepsilon_{i,t}.$$

The cross-price coefficients satisfy the interpretation

$$\gamma_{ij} = \frac{\partial Q_i^D}{\partial P_j}.$$

Positive γ_{ij} is consistent with substitution under the relevant price convention, while negative γ_{ij} is consistent with complementarity.

Because monetary prices and quantities are often jointly determined, instrumental variables, structural VARs, panel methods, state-space models, or simultaneous-equation estimators may be required.

25 Machine Learning and Artificial Intelligence

Machine learning can be used to estimate nonlinear demand systems.

Let

$$\hat{Q}_i^D = f_\theta(P_i, \mathbf{P}_{-i}, N_i, R_i, \sigma_i, X_t, G_t),$$

where f_θ is a learned function.

A supervised learning objective is

$$\min_{\theta} \sum_t (Q_{i,t} - f_\theta(\mathbf{X}_t))^2.$$

However, predictive accuracy alone is insufficient for economic inference.

A useful AI system should preserve:

1. economic constraints;
2. budget constraints;
3. no-arbitrage conditions;
4. market-clearing conditions;
5. monotonicity where economically justified;
6. positivity constraints;
7. causal identification where inference is desired.

A physics-informed or economics-informed neural architecture can therefore be represented as

$$\hat{\mathbf{Q}} = f_\theta(\mathbf{X}) \quad \text{subject to} \quad \mathbf{Q}^D = \mathbf{Q}^S.$$

26 International Monetary Competition

Fiat currencies compete internationally as well as domestically.

Let F_k denote the currency of country k .

Then

$$\mathcal{F} = \{F_1, \dots, F_K\}.$$

Each currency provides monetary services conditional upon:

- fiscal credibility;
- monetary credibility;
- capital-market depth;
- legal institutions;
- geopolitical influence;
- military capability;

- trade networks;
- payment infrastructure.

The international monetary system therefore contains a network structure.

Let A_{ij} denote the strength of monetary linkage between jurisdictions i and j .

Then

$$A = [A_{ij}]$$

is a monetary network adjacency matrix.

27 Geopolitics and Diplomacy

The utility of a monetary asset can depend on geopolitical exposure.

Write

$$V_i = V_i^{economic} + V_i^{institutional} + V_i^{geopolitical}.$$

A geopolitical shock G_t can therefore generate

$$\frac{\partial V_i}{\partial G_t} \neq 0.$$

Sanctions, capital controls, reserve freezes, payment restrictions, alliance structures, trade embargoes, and diplomatic relations can affect the relative attractiveness of monetary assets.

This does not imply that any alternative asset automatically defeats fiat. It means that the institutional environment enters the monetary demand system.

28 Warfare Economics

War changes the economics of money through several channels.

Let

$$\Omega_t$$

denote the state of geopolitical conflict.

Then

$$V_i = V_i(\Omega_t).$$

War can affect:

- physical accessibility;
- banking infrastructure;
- payment networks;
- commodity supply;
- energy prices;
- sanctions;
- capital controls;
- sovereign creditworthiness;
- cyber infrastructure.

Physical commodities may benefit from strategic-reserve demand, while digital assets may benefit from portability but suffer from technological, regulatory, or infrastructure risks.

The effect is therefore theoretically ambiguous.

29 Strategic Reserves

A sovereign may choose reserves according to

$$\max_{\mathbf{R}} U(\mathbf{R})$$

subject to

$$\sum_i P_i R_i \leq B.$$

The optimal reserve vector is

$$\mathbf{R}^* = (R_1^*, \dots, R_n^*).$$

This establishes a direct connection between monetary theory and national security.

Gold reserves, foreign exchange reserves, strategic commodity inventories, and potentially digital assets can therefore be analyzed within the same portfolio framework.

30 Political Economy

Fiat money is not merely an economic technology. It is also an institutional technology.

The state obtains benefits from monetary sovereignty, including:

- taxation in domestic currency;
- seigniorage;
- financial regulation;
- macroeconomic stabilization;
- emergency liquidity provision;
- sanctions enforcement;
- payment-system oversight.

The political economy of fiat therefore modifies its utility:

$$V_F = V_F^{private} + V_F^{state}.$$

Cryptocurrencies may reduce some forms of state control while increasing other forms of private coordination.

Commodity money occupies another institutional position because its physical existence is generally independent of a particular digital protocol.

31 Welfare Economics

Let social welfare be

$$W = \sum_{h=1}^H u_h(C_h, \mathbf{M}_h).$$

The competitive monetary equilibrium may not maximize welfare because of:

- network externalities;
- transaction externalities;
- information asymmetries;
- environmental externalities;

- systemic financial risk;
- monopoly or oligopoly power;
- security externalities.

Let the social marginal benefit of monetary asset i be

$$SMB_i = PMB_i + EMB_i,$$

where EMB_i is the external marginal benefit.
Similarly,

$$SMC_i = PMC_i + EMC_i.$$

The social optimum satisfies

$$SMB_i = SMC_i.$$

The market equilibrium need not satisfy this condition.

32 Environmental and Energy Economics

Different monetary technologies consume different physical resources.
For cryptocurrency mining or validation,

$$E_C = E_C(H, \mathcal{A}, \mathcal{T}),$$

where H denotes computational capacity.
For commodity extraction,

$$E_G = E_G(X, \text{ore grade, technology}).$$

For fiat systems, physical resource requirements arise primarily through banking, payment, communications, data centers, and administrative infrastructure.

The appropriate comparison is therefore not simply energy per transaction. A complete welfare analysis must consider the total monetary service delivered per unit of resource consumption.

33 A Unified Monetary Production Function

Define the monetary-service production function

$$Y_i = F_i(K_i, L_i, E_i, T_i, N_i, I_i),$$

where:

- K_i = capital;
- L_i = labor;
- E_i = energy;
- T_i = technology;
- N_i = network scale;
- I_i = institutional capital.

This permits the comparison of apparently unrelated monetary systems.

Gold mining produces a physical monetary asset.

Blockchain validation produces digital settlement and scarcity services.

Central banking and commercial banking produce institutional liquidity, settlement, and credit services.

The objects differ physically but can be compared economically through the monetary services they provide.

34 A General Monetary Production Frontier

Let monetary output be represented by

$$\mathbf{Y} = (Y_F, Y_C, Y_G, \dots).$$

Given resources $\mathbf{R}$, the feasible set is

$$\mathcal{F}(\mathbf{R}) = \{\mathbf{Y} : \mathbf{Y} \text{ is technologically feasible}\}.$$

The monetary system operates on or inside a generalized production possibilities frontier. Technological innovation shifts the frontier.

For example:

better blockchain technology $\rightarrow$ lower cost of crypto monetary services,

while

better extraction technology $\rightarrow$ greater feasible commodity supply.

35 Existence of Monetary Equilibrium

We now state a general equilibrium result.

Assumption 1. *For each monetary asset i , demand and supply are continuous functions of relative prices and aggregate state variables.*

Assumption 2. *The feasible portfolio set is nonempty, compact, and convex.*

Assumption 3. *Aggregate excess demand satisfies the appropriate Walrasian budget restriction.*

Theorem 1 (Existence of Monetary Equilibrium). *Under the preceding assumptions, there exists at least one vector of relative monetary prices $\mathbf{P}^*$ satisfying market clearing.*

Proof. Normalize one monetary asset as the numeraire. The remaining price vector belongs to a compact normalized price simplex. Continuity of demand and supply implies continuity of excess demand. Under the Walrasian budget restriction, the excess-demand correspondence maps the feasible price simplex into itself. Brouwer's fixed-point theorem therefore guarantees a fixed point $\mathbf{P}^*$ at which aggregate excess demand vanishes. $\square$ $\square$

The theorem establishes existence, not uniqueness.

36 Uniqueness and Stability

Let

$$\mathbf{Z}(\mathbf{P}) = \mathbf{Q}^D(\mathbf{P}) - \mathbf{Q}^S(\mathbf{P}).$$

A price-adjustment process may be represented as

$$\dot{P}_i = \lambda_i P_i Z_i(\mathbf{P}), \quad \lambda_i > 0.$$

An equilibrium is locally stable if the Jacobian

$$J = \frac{\partial \mathbf{Z}}{\partial \mathbf{P}}$$

satisfies appropriate eigenvalue restrictions.

If all relevant eigenvalues have negative real parts under the normalized system, local convergence follows.

Thus a monetary equilibrium may exist without being dynamically stable.

37 Multiple Equilibria

Network effects can generate multiple equilibria.

Suppose

$$V_C = a + bN_C - \theta P_C,$$

with adoption

$$N_C = N \frac{e^{V_C}}{e^{V_C} + e^{V_F} + e^{V_G}}.$$

A sufficiently large b can generate multiple fixed points.

The economy may then possess:

1. a low-crypto-adoption equilibrium;
2. an intermediate equilibrium;
3. a high-crypto-adoption equilibrium.

The same mathematical mechanism can apply to fiat currencies, gold, or other monetary networks.

38 Monetary Competition as an Evolutionary Process

Let s_i denote the market share of monetary technology i .

Then

$$\dot{s}_i = s_i (V_i - \bar{V}),$$

where

$$\bar{V} = \sum_j s_j V_j.$$

A monetary technology expands when its utility exceeds the population average.

This produces a generalized evolutionary monetary equation:

$$\dot{s}_i = s_i (V_i - \bar{V}).$$

The equation captures competition without assuming that the system necessarily converges to one dominant currency.

39 Monetary Dominance

Define the monetary dominance index

$$D_i = \omega_1 s_i + \omega_2 N_i + \omega_3 L_i + \omega_4 T_i + \omega_5 I_i,$$

with

$$\sum_k \omega_k = 1.$$

The asset with the largest D_i is the dominant monetary technology under the selected weighting scheme.

Dominance need not mean exclusive use.

A monetary system may instead be characterized by coexistence:

$$s_F > 0, \quad s_C > 0, \quad s_G > 0.$$

This is likely to be the empirically relevant case.

40 The Monetary Triangle

Figure 1 summarizes the fundamental competitive structure.

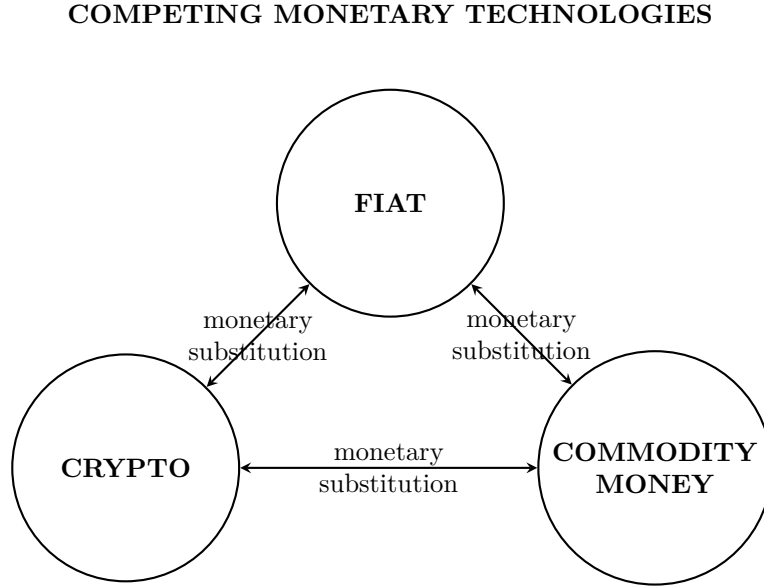


Figure 1: The fundamental monetary competition triangle.

41 The Broader Monetary Network

The three-node triangle can be generalized to many monetary assets.

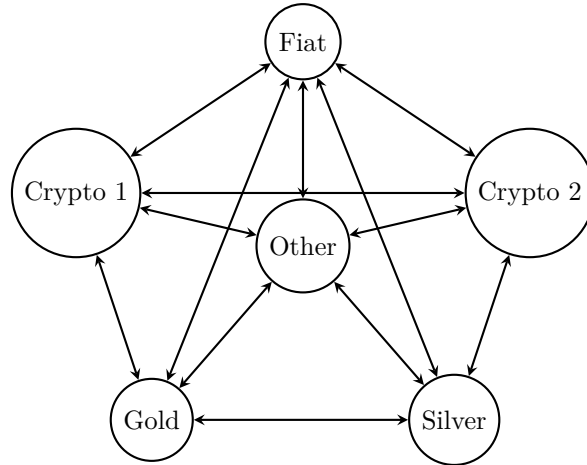


Figure 2: A generalized network of competing monetary assets.

Each edge represents a potential substitution, complementarity, arbitrage, or information channel.

42 Arbitrage and No-Arbitrage

For three assets i, j, k ,

$$P_{ij}P_{jk}P_{ki} = 1$$

must hold in a frictionless no-arbitrage environment.

With transaction costs τ_{ij} , arbitrage is profitable only if

$$P_{ij}P_{jk}P_{ki} > 1 + \tau$$

or

$$P_{ij}P_{jk}P_{ki} < \frac{1}{1 + \tau}.$$

Thus observed deviations from triangular parity can contain information about:

- transaction costs;
- liquidity;
- capital controls;
- settlement delays;
- market fragmentation;
- technological frictions.

43 Liquidity

Define market liquidity as a function

$$L_i = L_i(\text{depth, turnover, bid-ask spread, settlement speed, market-making capacity}).$$

A monetary asset with low liquidity may require a liquidity premium.

Let

$$\lambda_i$$

denote the liquidity discount.

Then perceived monetary utility may be

$$V_i^* = V_i - \lambda_i.$$

Liquidity can therefore become an endogenous component of monetary demand.

44 Collateral and Financial Intermediation

A monetary asset may have value because it can be pledged as collateral.

Let collateral capacity be

$$K_i = \chi_i P_i Q_i,$$

where χ_i is the collateralization ratio.

The total financial capacity of the monetary system is therefore

$$K = \sum_i K_i.$$

A change in the price of one monetary asset can consequently affect credit creation elsewhere in the financial system.

This provides a bridge between monetary competition and systemic financial risk.

45 Systemic Risk

Let w_i be the portfolio share of monetary asset i .
Portfolio variance is

$$\sigma_P^2 = \sum_i \sum_j w_i w_j \sigma_{ij}.$$

Diversification across monetary technologies can reduce risk if

$$\sigma_{ij} < \sigma_i \sigma_j$$

in the appropriate correlation representation.

However, common shocks can make apparently distinct monetary assets correlated.

For example:

global liquidity shock $\rightarrow$ fiat $\rightarrow$ crypto $\rightarrow$ commodity

through common portfolio rebalancing.

46 Strategic and Security Dimensions

Monetary assets also differ in their vulnerability to physical and digital attack.

Define resilience

$$R_i = R_i(\text{redundancy, decentralization, physical security, cybersecurity, institutional capacity}).$$

A monetary system exposed to a single point of failure has lower resilience than a sufficiently redundant system, other things equal.

This creates a monetary-security tradeoff:

$$\text{efficiency} \leftrightarrow \text{resilience}.$$

The tradeoff is relevant in peacetime as well as wartime.

47 Cybersecurity and Consensus

For decentralized digital monetary systems, security can be represented schematically as

$$S_C = S_C(H, q, d, r),$$

where:

- H is computational or staking capacity;
- q is attacker capacity;
- d is decentralization;
- r is redundancy.

A protocol's monetary utility depends partly on the credibility of its settlement mechanism.

Thus monetary demand can depend on computer science variables that have no direct analogue in physical commodity money.

48 A General Demand System

For n monetary assets, define

$$\mathbf{Q}^D = \mathbf{D}(\mathbf{P}, \mathbf{R}, \mathbf{N}, \mathbf{L}, \mathbf{I}, \mathbf{G}, \mathbf{X}).$$

The Jacobian is

$$J_D = \frac{\partial \mathbf{Q}^D}{\partial \mathbf{P}}.$$

Its off-diagonal entries

$$\frac{\partial Q_i^D}{\partial P_j}$$

measure substitution or complementarity.

Its diagonal entries

$$\frac{\partial Q_i^D}{\partial P_i}$$

capture own-price effects.

The Slutsky decomposition can then be generalized to monetary assets, separating substitution and wealth effects.

49 Supply Elasticity

Define supply elasticity for monetary asset i as

$$\varepsilon_i^S = \frac{\partial Q_i^S}{\partial P_i} \frac{P_i}{Q_i^S}.$$

Different monetary technologies can have radically different elasticities.

A protocol with a fixed issuance rule may have

$$\varepsilon_C^S \approx 0$$

over a sufficiently short horizon.

Physical commodity supply may also be highly inelastic in the short run:

$$\varepsilon_G^S \ll 1.$$

Fiat supply can be comparatively flexible because monetary authorities and banks can change nominal monetary conditions.

Thus the incidence of monetary demand shocks differs across monetary technologies.

50 A Fundamental Monetary Shock

Suppose demand for cryptocurrency increases:

$$\Delta Q_C^D > 0.$$

If

$$\varepsilon_C^S \approx 0,$$

the short-run adjustment may occur primarily through price:

$$\Delta P_C > 0.$$

But if crypto competes with gold and fiat, the full general-equilibrium effect is

$$\Delta s_C > 0, \quad \Delta s_F < 0, \quad \Delta s_G < 0,$$

provided the substitution effects dominate.

The same shock can therefore affect several monetary markets simultaneously.

51 Comparative Statics

Let equilibrium satisfy

$$\mathbf{Z}(\mathbf{P}, \mathbf{X}) = 0.$$

Differentiating,

$$\frac{\partial \mathbf{Z}}{\partial \mathbf{P}} d\mathbf{P} + \frac{\partial \mathbf{Z}}{\partial \mathbf{X}} d\mathbf{X} = 0.$$

Therefore,

$$d\mathbf{P} = - \left(\frac{\partial \mathbf{Z}}{\partial \mathbf{P}} \right)^{-1} \frac{\partial \mathbf{Z}}{\partial \mathbf{X}} d\mathbf{X}.$$

This is the central comparative-statics equation of the theory.

It allows monetary shocks, technological shocks, geopolitical shocks, inflation shocks, and network shocks to propagate through the complete monetary system.

52 Empirical Predictions

The theory generates several testable predictions.

1. A cryptocurrency's fiat price should depend partly on the relative attractiveness of fiat itself.
2. Crypto demand should respond to expected fiat inflation after controlling for crypto-specific risk.
3. Crypto demand should respond to the attractiveness of alternative commodity monies.
4. Gold demand should respond not only to industrial and jewelry demand but also to competition from other monetary assets.
5. Network adoption should affect cryptocurrency demand nonlinearly.
6. Monetary assets with stronger institutional credibility should command greater monetary portfolio shares, conditional on other characteristics.
7. Geopolitical shocks should alter relative demand across monetary technologies.
8. Strategic-reserve accumulation should affect equilibrium prices when the affected market is sufficiently inelastic.
9. Cross-price elasticities should differ substantially across monetary asset classes.
10. The volatility of an asset should affect its monetary demand even when its expected return is unchanged.

53 Identification Strategy

An empirical implementation can estimate

$$Q_{i,t} = \alpha_i + \beta_i P_{i,t} + \sum_{j \neq i} \gamma_{ij} P_{j,t} + \delta_i N_{i,t} + \eta_i R_{i,t} + \theta_i G_t + u_{i,t}.$$

Potential instruments include:

- protocol-driven supply changes;
- exogenous mining or staking changes;
- geological supply disruptions;
- monetary-policy surprises;

- exogenous geopolitical events;
- technological shocks;
- regulatory announcements.

The identification strategy must distinguish genuine causal effects from simultaneous market responses.

54 Data Architecture

A complete empirical database should combine:

Table 1: Suggested empirical data architecture		
Category	Variables	Examples
Fiat	money, inflation, rates	CPI, M1, M2, policy rate
Crypto	prices, supply, activity	market cap, volume, users
Commodity	prices, inventories	gold, silver, copper, platinum
Finance	risk and liquidity	yields, spreads, volatility
Network	adoption	addresses, transactions, developers
Geopolitics	conflict and sanctions	events, restrictions, trade
Energy	resource constraints	electricity, mining costs
Macro	real economy	GDP, employment, trade

The dataset should be constructed at multiple frequencies because monetary markets contain both high-frequency price formation and low-frequency institutional changes.

55 Machine-Learning Research Pipeline

A research pipeline may be organized as:

Raw Data $\rightarrow$ Cleaning $\rightarrow$ Feature Engineering $\rightarrow$ Economic Constraints $\rightarrow$ Estimation $\rightarrow$ Validation $\rightarrow$ Equilibrium Simulation.

Features may include:

$X_t = (\text{inflation, rates, volatility, network activity, commodity prices, geopolitical state, liquidity}).$

A hybrid model can then estimate

$$\hat{\mathbf{Q}}_t = f_{\theta}(\mathbf{X}_t)$$

while enforcing

$$\hat{\mathbf{Q}}_t^D - \hat{\mathbf{Q}}_t^S \approx 0.$$

56 Artificial Intelligence and Agent-Based Monetary Systems

An agent-based model can contain:

- households;
- firms;
- banks;
- central banks;
- cryptocurrency validators;

- miners;
- commodity producers;
- institutional investors;
- sovereigns;
- strategic actors.

Each agent h solves

$$\max_{\mathbf{M}_h} E_h \sum_t \beta_h^t u_h(C_{h,t}, \mathbf{M}_{h,t})$$

subject to its budget and technological constraints.

The resulting decentralized system can be simulated using Monte Carlo, agent-based computation, reinforcement learning, or multi-agent reinforcement learning.

57 Pseudo-Code for Agent-Based Monetary Equilibrium

INITIALIZE:

```
agents
monetary assets
prices
technologies
institutions
geopolitical state
```

FOR $t = 1, \dots, T$:

FOR each agent h :

```
observe:
  prices
  returns
  liquidity
  network states
  institutional information
  geopolitical information
```

```
estimate:
  expected returns
  risk
  monetary utility
```

```
choose:
  fiat holdings
  crypto holdings
  commodity holdings
  other asset holdings
```

AGGREGATE:

```
monetary demand
transaction demand
collateral demand
investment demand
```

COMPUTE:

```
protocol issuance
commodity production
```

```

    fiat creation
    effective market supply

```

```

CLEAR MARKETS:
    update relative prices

```

```

UPDATE:
    networks
    beliefs
    institutions
    geopolitical state
    technology

```

```

RECORD:
    prices
    quantities
    market shares
    volatility
    welfare
    systemic risk

```

```

RETURN:
    equilibrium distributions
    transition dynamics
    stress-test outcomes

```

58 Stress Testing

The framework permits monetary stress testing.
Consider shocks:

$$S = \left\{ \begin{array}{l} \text{inflation} \text{ war} \text{ sanctions} \text{ banking} \text{ commodity} \\ \text{shock} \text{ 'shock'} \text{ shock} \text{ , crisis} \text{ , cyberattack, protocol , supply} \\ \text{shock} \text{ failure , shock} \end{array} \right\}.$$

For each shock s ,

$$\mathbf{P}_s^* = \text{Equilibrium}(\mathbf{X} + s).$$

The difference

$$\Delta \mathbf{P}_s = \mathbf{P}_s^* - \mathbf{P}_0^*$$

measures the equilibrium response.

This makes the theory directly applicable to financial stability and national-security analysis.

59 Welfare and Policy

A policymaker should not automatically maximize the market share of one monetary technology.
Instead, the objective may be

$$\max [W - \lambda_1 \text{Risk} - \lambda_2 \text{Externality} + \lambda_3 \text{Resilience}].$$

The optimal policy can involve coexistence rather than monopoly.

Possible policy instruments include:

- monetary policy;
- prudential regulation;
- reserve requirements;

- capital requirements;
- commodity reserves;
- payment-system regulation;
- taxation;
- disclosure;
- cybersecurity standards;
- settlement infrastructure.

The framework does not prescribe one particular policy regime. It provides the structure for evaluating alternative regimes.

60 Philosophical Interpretation

The theory has an underlying philosophical implication.

Money is not necessarily a thing.

It can be understood as a ****socially coordinated technology for transferring purchasing power through time and across agents****.

A physical metal can perform this function.

A government liability can perform this function.

A cryptographic token can perform this function.

A bank deposit can perform this function.

The common economic property is therefore not material composition but monetary service.

We may write

Money = technology for coordinating claims on scarce resources.

The different monetary technologies implement that function through different institutional, physical, computational, and political mechanisms.

61 Political and Geopolitical Interpretation

Monetary systems are also systems of power.

A currency with widespread international use can generate:

- seigniorage;
- financial influence;
- sanctions capacity;
- payment-system power;
- diplomatic leverage;
- borrowing advantages;
- reserve accumulation;
- strategic autonomy.

Consequently, competition among monetary assets can become competition among political institutions.

The international monetary system can therefore be viewed as a network game:

$$\max_i U_i(\text{economic welfare, monetary sovereignty, financial security, geopolitical influence}).$$

This does not imply that monetary competition inevitably causes conflict. It means that monetary institutions can become strategic assets within international relations.

62 A General Theory of Monetary Substitution

The preceding results can be summarized in a single principle.

Theorem 2 (General Monetary Substitution Principle). *Suppose a finite set of monetary assets competes for a fixed aggregate portfolio of monetary wealth. If the utility of monetary asset i rises relative to asset j , and portfolio constraints permit substitution, then the equilibrium share of i weakly increases and the equilibrium share of at least one competing asset weakly decreases, subject to general-equilibrium feedback effects.*

Proof. Portfolio shares satisfy

$$\sum_i s_i = 1.$$

Suppose V_i increases while the other utilities remain unchanged. Under the multinomial allocation rule,

$$s_i = \frac{e^{V_i}}{\sum_j e^{V_j}},$$

so

$$\frac{\partial s_i}{\partial V_i} = s_i(1 - s_i) > 0.$$

For $j \neq i$,

$$\frac{\partial s_j}{\partial V_i} = -s_i s_j < 0.$$

Hence an increase in relative monetary utility reallocates portfolio share toward asset i and away from competing assets. General-equilibrium feedback may modify the magnitude and incidence of the effect but cannot eliminate the portfolio constraint itself. $\square$ $\square$

63 General Theory of Monetary Supply

The supply side can likewise be unified.

For each monetary technology i ,

$$S_i = S_i(P_i, \mathcal{R}_i, \mathcal{T}_i, \mathcal{I}_i, \mathcal{G}_i),$$

where:

- $\mathcal{R}_i$ represents physical resources;
- $\mathcal{T}_i$ represents technology;
- $\mathcal{I}_i$ represents institutions;
- $\mathcal{G}_i$ represents governance.

The three principal cases are:

$$S_F = S_F(\mathcal{I}_F, \mathcal{G}_F, \mathcal{T}_F),$$

$$S_C = S_C(\mathcal{T}_C, \mathcal{G}_C),$$

and

$$S_G = S_G(\mathcal{R}_G, \mathcal{T}_G).$$

This formulation allows radically different monetary systems to be analyzed within one mathematical framework.

64 Central Result

The complete theory can be represented by

$$\begin{aligned} Q_i^D &= D_i(\mathbf{P}, \mathbf{R}, \mathbf{N}, \mathbf{L}, \mathbf{I}, \mathbf{G}, \mathbf{X}), \\ Q_i^S &= S_i(\mathbf{P}, \mathbf{R}_i, \mathbf{T}_i, \mathbf{I}_i, \mathbf{G}_i), \\ Q_i^D &= Q_i^S, \quad i = 1, \dots, n. \end{aligned}$$

The equilibrium vector is

$$\mathbf{P}^* = (P_1^*, \dots, P_{n-1}^*)$$

after normalization.

The equilibrium is therefore jointly determined by:

1. preferences
2. expectations
3. technology
4. scarcity
5. institutions
6. networks
7. politics
8. geopolitics
9. physical constraints.

65 Conclusion

This paper has developed a general theory of demand and supply for monetary assets in which fiat currencies, cryptocurrencies, commodity monies, and other monetary technologies compete for scarce economic resources and portfolio wealth.

The fundamental departure from conventional analysis is conceptual.

A cryptocurrency quoted in dollars is not being valued against an economically neutral numeraire. It is being valued against another monetary asset that itself has demand, supply, institutional credibility, inflation characteristics, liquidity, political backing, and geopolitical utility.

Likewise, gold priced in dollars is not simply a commodity with a dollar price. Gold can itself function as a monetary asset competing with fiat, cryptocurrency, and other stores of value.

Consequently, the appropriate object of analysis is a system of relative monetary prices:

$$P_{ij} = \frac{P_i}{P_j}.$$

The monetary economy is therefore a competitive network rather than a single-market hierarchy.

The three principal monetary scarcity technologies are:

institutional scarcity (fiat)

algorithmic scarcity (cryptocurrency)

and

physical scarcity (commodity money).

Their competition occurs through a common set of monetary functions: transaction, settlement, liquidity, collateral, store of value, accounting, and preservation of purchasing power.

The resulting theory therefore transforms the question from

“What determines the price of a cryptocurrency?”

into

“What determines the equilibrium allocation of monetary wealth among competing monetary technologies?”

This formulation provides a unified analytical foundation for studying fiat currencies, cryptocurrencies, gold, silver, platinum, copper, digital money, strategic reserves, monetary sovereignty, financial stability, geopolitical monetary competition, and future monetary technologies within one general-equilibrium framework.

Notation

Table 2: Principal notation

Symbol	Meaning
$\mathcal{M}$	Set of monetary assets
F	Fiat currency class
C	Cryptocurrency class
G	Commodity-money class
P_i	Price of monetary asset i
P_{ij}	Relative price of asset i in units of j
Q_i^D	Demand for monetary asset i
Q_i^S	Supply of monetary asset i
s_i	Portfolio share of monetary asset i
V_i	Monetary utility of asset i
N_i	Network size of asset i
L_i	Liquidity of asset i
R_i	Expected return on asset i
σ_i	Risk of asset i
I_i	Institutional credibility
G_i	Geopolitical characteristics
W	Aggregate monetary wealth
X_t	Macroeconomic state vector

References

- [1] K. J. Arrow and G. Debreu, “Existence of an Equilibrium for a Competitive Economy,” *Econometrica*, Vol. 22, 1954, pp. 265–290.
- [2] L. Walras, *Elements of Pure Economics*, 1874.
- [3] J. M. Keynes, *The General Theory of Employment, Interest and Money*, Macmillan, 1936.
- [4] M. Friedman, “The Quantity Theory of Money: A Restatement,” in *Studies in the Quantity Theory of Money*, University of Chicago Press, 1956.
- [5] R. E. Lucas, “Econometric Policy Evaluation: A Critique,” Carnegie-Rochester Conference Series on Public Policy, Vol. 1, 1976, pp. 19–46.
- [6] P. A. Samuelson, *Economics*, McGraw-Hill, 1948.
- [7] K. J. Arrow, “An Extension of the Basic Theorems of Classical Welfare Economics,” in *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*, University of California Press, 1951.

- [8] F. A. Hayek, “The Use of Knowledge in Society,” *American Economic Review*, Vol. 35, 1945, pp. 519–530.
- [9] C. Menger, “On the Origin of Money,” *Economic Journal*, Vol. 2, 1892, pp. 239–255.
- [10] E. F. Fama, “Efficient Capital Markets: A Review of Theory and Empirical Work,” *Journal of Finance*, Vol. 25, 1970, pp. 383–417.
- [11] H. Markowitz, “Portfolio Selection,” *Journal of Finance*, Vol. 7, 1952, pp. 77–91.
- [12] W. F. Sharpe, “Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk,” *Journal of Finance*, Vol. 19, 1964, pp. 425–442.
- [13] G. A. Akerlof, “The Market for ‘Lemons’: Quality Uncertainty and the Market Mechanism,” *Quarterly Journal of Economics*, Vol. 84, 1970, pp. 488–500.
- [14] D. Kahneman and A. Tversky, “Prospect Theory: An Analysis of Decision under Risk,” *Econometrica*, Vol. 47, 1979, pp. 263–291.
- [15] C. E. Shannon, “A Mathematical Theory of Communication,” *Bell System Technical Journal*, Vol. 27, 1948, pp. 379–423 and 623–656.
- [16] S. Nakamoto, “Bitcoin: A Peer-to-Peer Electronic Cash System,” 2008.
- [17] V. Buterin, “A Next-Generation Smart Contract and Decentralized Application Platform,” Ethereum White Paper, 2014.
- [18] F. A. Hayek, *The Denationalization of Money*, Institute of Economic Affairs, 1976.
- [19] E. Ostrom, *Governing the Commons*, Cambridge University Press, 1990.
- [20] W. B. Arthur, “Competing Technologies, Increasing Returns, and Lock-In by Historical Events,” *Economic Journal*, Vol. 99, 1989, pp. 116–131.
- [21] W. B. Arthur, *Increasing Returns and Path Dependence in the Economy*, University of Michigan Press, 1994.
- [22] R. M. Metcalfe, “Metcalfe’s Law after 40 Years of Ethernet,” *Computer Networks and ISDN Systems*, 1995.
- [23] L. von Mises, *The Theory of Money and Credit*, 1912.
- [24] C. Menger, *Principles of Economics*, 1871.
- [25] M. Friedman and A. J. Schwartz, *A Monetary History of the United States, 1867–1960*, Princeton University Press, 1963.
- [26] R. E. Lucas, *Studies in Business-Cycle Theory*, MIT Press, 1981.
- [27] P. A. Samuelson and W. D. Nordhaus, *Economics*, McGraw-Hill.
- [28] Bank for International Settlements, *Annual Economic Report*, Bank for International Settlements.
- [29] International Monetary Fund, *Global Financial Stability Report*, International Monetary Fund.

Glossary

Algorithmic Scarcity

Scarcity generated by explicit computational or protocol rules governing issuance and supply.

Commodity Money

A physical commodity that provides monetary services in addition to any industrial, cultural, or consumption uses.

Cryptocurrency

A digitally native monetary or financial asset whose ownership and transactions are maintained through cryptographic and computational infrastructure.

Effective Market Supply

The quantity of an asset actually available for exchange at prevailing prices.

Fiat Money

Money whose value derives substantially from institutional, legal, fiscal, monetary, and network arrangements rather than convertibility into a specified physical commodity.

Institutional Scarcity

Scarcity determined by political, legal, monetary, or institutional rules.

Liquidity

The ability to transact in an asset rapidly and at low cost without causing large price changes.

Monetary Asset

An asset providing one or more monetary services.

Monetary Competition

Competition among monetary assets for transaction, savings, settlement, collateral, liquidity, and store-of-value demand.

Monetary Network

A network of agents, institutions, technologies, and payment relationships supporting a monetary asset.

Monetary Substitution

The reallocation of wealth or monetary demand from one monetary asset toward another.

Physical Scarcity

Scarcity generated by finite physical resources and the technological constraints governing their extraction, transformation, and recycling.

Protocol Supply

The quantity of cryptocurrency units permitted by the protocol's consensus and issuance rules.

Relative Monetary Price

The price of one monetary asset expressed in units of another monetary asset.

Strategic Reserve

A stock of monetary, commodity, energy, or other economically important assets maintained to increase resilience against future shocks.

Network Effect

A mechanism whereby the utility of an asset changes with the size or structure of its user network.

Monetary Dominance

The relative importance of a monetary technology in the provision of monetary services.

Monetary Technology

The institutional, physical, computational, and organizational technology through which monetary services are produced.

Seigniorage

Economic income associated with the ability to issue monetary liabilities.

Store of Value

The ability of an asset to transfer purchasing power through time.

The End